

The Multidimensional Geometry of Truth in Large Language Models: A Concept-Cone Extension of the Linear Representation Hypothesis

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Abstract

The Linear Representation Hypothesis holds that high-level concepts are encoded as linear directions in a language model’s residual stream. Under this hypothesis, propositional truth is modeled as one such direction: a Difference-in-Means (DIM) probe recovers it, and adding or ablating that direction can flip the model’s True/False verdict. Recent work on refusal suggests this single direction may be the one-dimensional case of a multi-dimensional *concept cone*, a linear subspace whose interior mediates the behaviour. This project asks whether truth has the same structure. I test two methods: *Truth Direction Optimization* (TDO), a gradient-based refinement of the DIM direction under a loss that encodes two causal axioms and a KL-retention term; and *Truth Cone Optimization* (TCO), which extends this to a k -dimensional orthonormal cone and, at every step, samples directions from *inside* the cone so the optimizer targets the interior rather than only the basis. On six instruction-tuned models from three families (Qwen-2.5 1.5/7/14B, Gemma-2 2/9B, Llama-3.1-8B), a single direction is causally sufficient for the smaller models but fails for the larger ones on two axes at once: low ablation Answer Switching Rate and raised KL divergence on unrelated prompts. A 2-D cone closes both gaps for every model. The basis vectors are near-orthogonal to the DIM probe ($|\cos| < 0.1$) yet each is individually causally effective. For the models I tested, this is consistent with truth occupying a small multi-dimensional linear subspace rather than a single line.

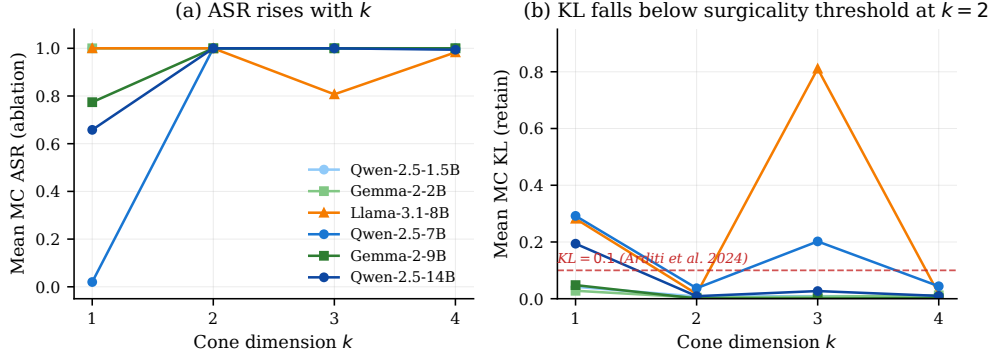


Fig. 1 $k=2$ resolves both axes simultaneously for larger models. (a) Mean Monte-Carlo Answer Switching Rate over 32 directions sampled from the learned cone (in-distribution ablation) versus cone dimension k . For smaller models (Qwen-2.5-1.5B, Gemma-2-2B, Llama-3.1-8B) a 1-dim direction already saturates; for Qwen-2.5-7B/14B and Gemma-2-9B a 1-dim direction is causally insufficient, but $k=2$ closes the gap. (b) Mean KL divergence on 100 retain prompts (last 30 token positions). The same three larger models that suffered low ASR at $k=1$ also *exceed* the surgicity threshold $KL = 0.1$ (red dashed line) at $k=1$, and *all* fall below it at $k=2$. Across these models, the $k=2$ cone is both more effective and less destructive than the 1-D probe. Llama-3.1-8B and Qwen-2.5-7B regress at $k=3$, identifying $k=2$ as the safe operating point.

1 Introduction

As large language models (LLMs) have scaled in size and capabilities, a sustained effort has emerged to render their internal computations interpretable. A central hypothesis in this effort, the *Linear Representation Hypothesis*, holds that high-level concepts such as sentiment, refusal, or factuality align with linear directions in a model’s residual stream, and that intervening along such a direction changes the corresponding behaviour (Park et al. 2024).

This work focuses on the concept of *propositional truth*. Given a binary factual statement (“The city of Tokyo is in Japan”), can the computation that decides between “True” and “False” be isolated and causally manipulated? This matters because reliable factual behaviour is needed before LLMs are used in high-stakes settings, and because understanding how truth is represented is a step toward predicting when it fails.

Earlier causal work treats truth as a single direction. Azaria and Mitchell (2023) showed that hidden-layer activations carry a truthfulness signal a shallow classifier can read, reporting 71–83% accuracy on binary factual statements (OPT-6.7B, LLaMA-2-7B). Marks and Tegmark (2024) made this geometric and causal: using PCA, cross-dataset probe generalization, and activation patching on LLaMA-2 (7B, 13B, 70B), they showed that truth representations emerge with scale, sit in a small group of hidden states above a “summarization” token, and that a Difference-in-Means (DIM) probe finds a direction more causally implicated in outputs than logistic-regression or contrastive alternatives. I take their layer-localization protocol and the DIM probe as my baseline, and extend it in two ways: from classification accuracy to causal mediation through a composite loss, and from one direction to a k -dimensional cone.

The concept cone, introduced by Wollschläger et al. (2025) for refusal, formalizes the multi-dimensional case. A concept W is represented by a concept cone of dimension

k if there is an orthonormal basis $V = [v_1, \dots, v_k]$ such that every direction in the polyhedral cone $\text{Cone}(V) = \{\sum_{i=1}^k \lambda_i v_i : \lambda_i \geq 0\} \setminus \{0\}$ mediates W in the sense of Definition 2.1. Each v_i is a linear direction, so the cone is a higher-dimensional *linear* object and the single-direction picture is its $k = 1$ case; the cone therefore extends the linear representation hypothesis to a multi-dimensional regime rather than replacing it.

This work asks two questions: does the concept-cone framework transfer from refusal to propositional truth, and if so, at what effective dimensionality k and at what cost to unrelated capabilities?

To answer them I make two methodological contributions and one empirical one. **(C-a)** TDO, a gradient-based 1-D refinement of DIM under a composite loss encoding two causal axioms (Definition 2.1) and a KL-retention term. **(C-b)** TCO, which extends TDO to k -dimensional orthonormal cones via modified Gram–Schmidt and Monte-Carlo sampling of interior directions during training, so the optimizer constrains the cone interior, not only its basis. **(C-c)** an evaluation across three families and scales (Qwen-2.5 1.5/7/14B, Gemma-2 2/9B, Llama-3.1-8B). A 1-D direction is causally sufficient for the smaller models (≤ 2 B) but fails for the three larger ones on two axes at once: low ablation Answer Switching Rate and raised KL divergence on unrelated prompts. A 2-D cone closes both gaps (Figure 1); and in the reference 4-D cone every basis vector, including the warm-started v_1 , is near-orthogonal to DIM ($|\cos| < 0.1$) yet the cone remains causally effective (Table 4), so its mediating directions are ones the linear probe does not recover.

I keep the scope narrow. The study covers instruction-tuned models only; forced-choice True/False prompts over three domains (geography, biology, chemistry); an 80/20 in-distribution split; and a leave-one-domain-out out-of-distribution test. Open-ended generation, multi-hop reasoning, base models, and adversarial robustness of the cone are left to future work (Section 4).

2 Method

I study factual knowledge as a forced binary choice: the model is shown a factual statement and must decide whether it is “True” or “False”. Given a frozen instruction-tuned model f , my goal is to find a subspace of f ’s residual stream that causally controls this decision, a set of directions whose manipulation flips the model’s verdict.

Section 2.1 states the two causal properties a candidate truth direction must satisfy; Section 2.2 describes the activation-patching step that selects a target layer l^* per model; Section 2.3 gives the DIM baseline and the magnitude calibration $\alpha = \|\theta_{\text{DIM}}\|_2$; Section 2.4 introduces TDO; Section 2.5 extends it to the k -dimensional cone; and Section 2.6 defines the evaluation protocols.

Factual evaluation.

Concretely, following Marks and Tegmark (2024), each prompt uses a few-shot, in-context format rather than a conversational one:

```
The city of Tokyo is in Japan. This statement is: True
The city of Hanoi is in Poland. This statement is: False
The city of Chicago is in Canada. This statement is:
```

The model’s next-token distribution is restricted to the two token IDs $t_{\text{true}} = \text{“True”}$ and $t_{\text{false}} = \text{“False”}$. The fixed suffix “This statement is:” separates the adverbial phrase and the closing punctuation, which together carry the truth-relevant edit, from the position at which the final layer reads out the prediction; this separation lets the activation-patching step (Section 2.2) isolate the truth signal before readout. The few-shot format also reduces interference from instruction-following circuits introduced during RLHF.

The training data are triples $(p_{\text{true}}, p_{\text{false}}, p_{\text{retain}})$: two factual statements with opposite truth value and one filtered Alpaca instruction with no factual-recall keywords.

2.1 Truth mediation properties

Let f be a frozen autoregressive transformer with $L \in \mathbb{N}$ layers and hidden size $d \in \mathbb{N}$, and let $x_i^{(l)}(p) \in \mathbb{R}^d$ denote its residual-stream activation at layer l and token position i on prompt p . A candidate truth direction is a unit vector $r \in \mathbb{S}^{d-1} = \{v \in \mathbb{R}^d : \|v\|_2 = 1\}$; its multi-dimensional generalization is a vector $r \in \text{Cone}(V)$ drawn from the concept cone $\text{Cone}(V) = \{\sum_{i=1}^k \lambda_i v_i : \lambda_i \geq 0\} \setminus \{0\}$ spanned by an orthonormal basis $V = [v_1, \dots, v_k] \in \mathbb{R}^{d \times k}$, of which the single-direction case is $k = 1$. Its causal role is probed by intervening on the activations $\{x_i^{(l)}(p)\}_{l,i}$ through the two operations below, and r , or, more generally, a concept cone $\text{Cone}(V) \subseteq \mathbb{R}^d$ of such directions, is admissible only if these interventions certify the two causal axioms of Definition 2.1.

Intervention operations.

Directional ablation $f_{\text{ablate}(r)}$ projects the stream onto the subspace orthogonal to r at every layer and position:

$$\forall i \forall l \quad \tilde{x}_i^{(l)} = x_i^{(l)} - \hat{r} \hat{r}^\top x_i^{(l)}, \quad (1)$$

where $\hat{r}$ is the unit vector along r . If the verdict flips from “True” to “False” after ablation, r was causally necessary.

Activation addition $f_{\text{add}(r, l^*, \alpha)}(p_{\text{false}})$ injects a scaled truth signal at a single layer l^* :

$$\forall i \quad \tilde{x}_i^{(l^*)} = x_i^{(l^*)} + \alpha \cdot r. \quad (2)$$

If a false statement flips to “True” under addition, r was causally sufficient to override the original judgment. Following [Arditi et al. \(2024\)](#), ablation is applied across all layers and token positions, and addition at the single selected layer l^* (Section 2.2).

Definition 2.1 (Truth mediation). *A direction $r \in \mathbb{R}^d$ at layer l^* mediates truth if:*

1. **Monotonic scaling.** *For every false statement p_{false} , the probability of “True” under $f_{\text{add}(r, l^*, \alpha)}(p_{\text{false}})$ increases monotonically with the coefficient $\alpha \in [0, \alpha_{\text{max}}]$.*
2. **Surgical factuality ablation.** *The directional ablation $f_{\text{ablate}(r)}(p_{\text{true}})$ satisfies (i) causal necessity: on true statements p_{true} , the model flips its judgment from “True” to “False” with high probability; and (ii) functional specificity: on retain*

prompts p_{retain} , the output distribution is preserved ($\text{KL} < 0.1$ per [Arditi et al. \(2024\)](#)).

A k -dim cone basis V mediates truth if every direction sampled from the relative cone $\text{Cone}(V)$ satisfies both axioms.

The cone version of Definition 2.1 motivates the Monte-Carlo sampling in Section 2.5: optimizing only on the basis vectors leaves the interior of $\text{Cone}(V)$ unconstrained, so it need not satisfy the k -dim version of the axioms.

Two metrics quantify these axioms and are reported throughout. The **Answer Switching Rate** (ASR) is the fraction of factual prompts whose argmax token flips under intervention: ASR_{abl} ablates p_{true} and counts flips to “False” (necessity); ASR_{add} adds to p_{false} and counts flips to “True” (sufficiency). **KL retention** $\text{KL}(p_r || p_0)$ is the divergence between the last-30-token distributions on p_{retain} with and without ablation of r (functional specificity).

2.2 Localizing the truth-mediating layer

Both TDO and TCO optimize at a single layer l^* , and running the full optimization at every layer is too expensive, so I first localize, per model, the layer whose residual-stream states most strongly mediate the truth judgment.

To do this I use denoising activation patching ([Heimersheim and Nanda 2024](#)), following [Marks and Tegmark \(2024\)](#). For each model I sample 50 contrastive pairs per dataset (cities, animals, elements). Each clean prompt p_{clean} is corrupted to p_{corrupt} by Symmetric Token Replacement (STR): the object token is swapped for a semantically related alternative that flips the ground-truth label while preserving syntactic structure and token count. I then cache the clean-run activations, re-run the model on p_{corrupt} , and restore a single activation at position i , layer l from the clean cache. To measure how much a layer contributes, I use the **logit difference** $M(p) = \text{logit}(p)_{t_{\text{true}}} - \text{logit}(p)_{t_{\text{false}}}$; following [Zhang and Nanda \(2024\)](#) I prefer it to a probability-based metric, since probability can miss components that contribute negatively to the correct answer, while the logit difference is linear in the final residual stream and so decomposes cleanly under patching. Writing M_{clean} , M_{corrupt} , and M_{patched} for the logit difference on the clean run, the corrupted run, and the restored run, the effect of that restoration is the normalized patching effect

$$\text{NPE}(l, i) = \frac{M_{\text{patched}} - M_{\text{corrupt}}}{M_{\text{clean}} - M_{\text{corrupt}}}, \quad (3)$$

which lies in $[0, 1]$: it is 1 when restoring (l, i) fully recovers clean performance and 0 at the corrupted baseline.

Sweeping NPE over every layer and position, then aggregating across models and datasets, reveals three groups of causally implicated hidden states, matching [Marks and Tegmark \(2024\)](#): group (a), the entity token that STR perturbs, encoding the factual association; group (b), the end-of-statement punctuation in mid layers, where clause-level information is aggregated into a propositional judgment; and group (c), the

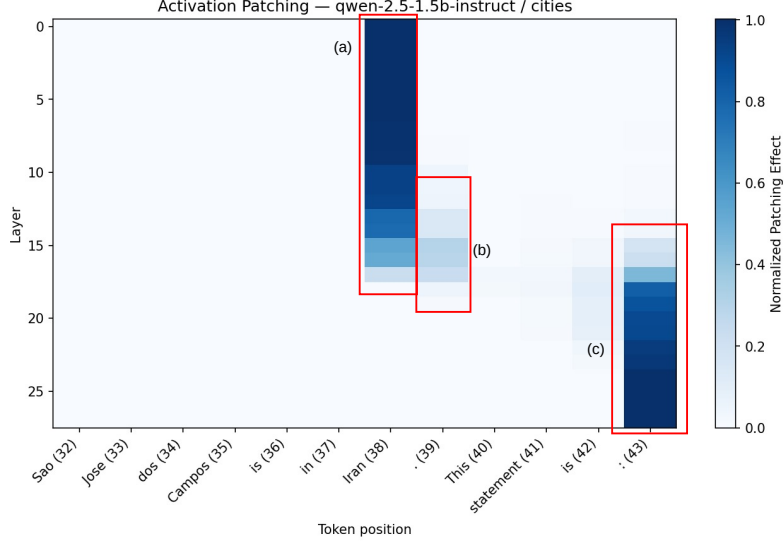


Fig. 2 Activation patching heatmap for Qwen-2.5-1.5B-Instruct on the cities domain, averaged over all 50 statement pairs.

colon token “:” in mid-to-late layers, which encodes the model’s prediction (applying the decoder head to these states gives top logits for “True”/“False”). I take l^* from group (b), the propositional judgment itself rather than its readout in (c), as the most-downstream layer there with mean NPE > 0.1 . Across all six models the selected layer falls at normalized depth 0.44–0.69 (Table 1).

This reduces the search space for all later optimization to a single $(l^*, -5)$ pair per model.¹ DIM, TDO, and TCO then all operate at the layer most causally implicated for that model, so the cross-family comparison does not rely on a fixed depth.

2.3 DIM baseline and magnitude calibration

A linear baseline both anchors the comparison and gives a starting point for optimization. Following Marks and Tegmark (2024); Arditi et al. (2024) I compute $\theta_{\text{DIM}} = \mu_+ - \mu_-$, where μ_+ and μ_- are the mean activations at $(l^*, -5)$ over the true and false training subsets. θ_{DIM} is normalized to a unit vector for ablation and used at its natural norm $\alpha := \|\theta_{\text{DIM}}\|_2$ for addition. Calibrating α to the model’s own representational scale at l^* , rather than tuning it by hand, removes one degree of freedom; per-model α ranges from 4.07 (Llama-3.1-8B) to 106.56 (Gemma-2-9B), reflecting large differences in residual-stream norm across families (Table 1).

¹The position $i = -5$ follows from the prompt structure: for all six tokenizers the four trailing tokens after the final word of the factual statement are the fixed suffix “This statement is:”, so position -5 lands on the end-of-statement punctuation.

2.4 Truth Direction Optimization (TDO)

DIM finds a direction that maximally separates the means of true and false activations, but mean separation does not guarantee causal mediation: a separating vector may fail to flip the model under ablation and may not preserve general capabilities. I therefore encode the causal axioms (Definition 2.1) directly into loss functions and recover a corresponding truth direction $r \in \mathbb{R}^d$ by gradient descent.

For the sufficiency property, I train the model to predict “True” on false statements p_{false} by running f with a modified forward pass $f_{\text{add}(r, l^*, \alpha)}(p_{\text{false}})$ that adds r to the activations at layer l^* with scaling coefficient α , minimizing the cross-entropy between the model output and the target token t_{true} . For the necessity property, I symmetrically compute the cross-entropy between the target token t_{false} and the output of a modified forward pass $f_{\text{ablate}(r)}$ that projects r out of every residual stream position, driving the model to predict “False” on true statements p_{true} . A key strength of the gradient-based approach is the ability to control any predefined objective, which lets me bound the extent to which unrelated computations are perturbed. For this, I use a retain loss based on the Kullback–Leibler (KL) divergence to ensure that directional ablation of r on retain prompts p_{retain} does not shift the model’s output distribution. The three terms combine into a composite loss

$$\mathcal{L}(r) = \lambda_{\text{abl}} \mathcal{L}_{\text{abl}}(r) + \lambda_{\text{add}} \mathcal{L}_{\text{add}}(r) + \lambda_{\text{ret}} \mathcal{L}_{\text{ret}}(r),$$

where

$$\begin{aligned} \mathcal{L}_{\text{abl}}(r) &= \text{CE}(f_{\text{ablate}(r)}(p_{\text{true}}), t_{\text{false}}) && \text{(necessity)}, \\ \mathcal{L}_{\text{add}}(r) &= \text{CE}(f_{\text{add}(r, l^*, \alpha)}(p_{\text{false}}), t_{\text{true}}) && \text{(sufficiency)}, \\ \mathcal{L}_{\text{ret}}(r) &= \text{KL}(f_{\text{ablate}(r)}(p_{\text{retain}}) \parallel f(p_{\text{retain}})) && \text{(surgicity)}. \end{aligned}$$

I initialize r from $\theta_{\text{DIM}} / \|\theta_{\text{DIM}}\|$ (warm start), placing the optimizer in the discriminative region rather than at a random point on the unit sphere. The loss weights $(\lambda_{\text{abl}}, \lambda_{\text{add}}, \lambda_{\text{ret}}) = (1.0, 0.2, 1.0)$ are taken from Wollschläger et al. (2025, Tab. 3) and held fixed across all six models (Table 6). Algorithm 1 gives the full procedure.

2.5 Truth Cone Optimization (TCO)

Table 2 shows that even TDO at $k = 1$ fails for the three larger models on ablation (e.g. Qwen-2.5-7B: ASR = 0.020). If truth occupies a multi-dimensional linear subspace, one direction can only capture a projection of it. I therefore extend TDO to higher dimensions by searching for regions in activation space where all vectors mediate the truth verdict. For this, I optimize an orthonormal basis $V = [v_1, \dots, v_k] \in \mathbb{R}^{d \times k}$ spanning a k -dimensional polyhedral cone $\text{Cone}(V) = \{\sum_{i=1}^k \lambda_i v_i \mid \lambda_i \geq 0\} \setminus \{0\}$, where every direction $r \in \text{Cone}(V)$ satisfies the causal axioms (Definition 2.1). The constraint $\lambda_i \geq 0$ ensures that directions within the cone consistently strengthen the truth signal; allowing negative coefficients could introduce opposing effects and reduce overall effectiveness. Orthogonality of the basis vectors prevents the optimizer from

Algorithm 1 Truth Direction Optimization (TDO)

Require: Frozen model f , scaling coefficient α , addition layer l^* , learning rate η , loss weights $\lambda_{\text{abl}}, \lambda_{\text{add}}, \lambda_{\text{ret}}$, and data $\mathcal{D} = \{(p_{\text{true},i}, p_{\text{false},i}, p_{\text{retain},i})\}_{i=1}^N$.

Ensure: Truth direction r

```
1:  $r \leftarrow \theta_{\text{DIM}} / \|\theta_{\text{DIM}}\|_2$  ▷ warm-start from DIM
2: while not converged do
3:   Sample batch  $\mathbf{B} \sim \mathcal{D}$ 
4:    $\mathcal{L} \leftarrow \text{COMPUTELOSS}(r, f, \mathbf{B})$ 
5:    $r \leftarrow r - \eta \nabla_r \mathcal{L}$ 
6:    $r \leftarrow r / \|r\|_2$  ▷ renormalize to unit sphere
7: end while
8:
9: function COMPUTELOSS( $r, f, \mathbf{B}$ )
10:    $p_{\text{true}}, p_{\text{false}}, p_{\text{retain}} \leftarrow \mathbf{B}$ 
11:    $\mathcal{L}_{\text{abl}} \leftarrow \text{CE}(f_{\text{ablate}(r)}(p_{\text{true}}), t_{\text{false}})$ 
12:    $\mathcal{L}_{\text{add}} \leftarrow \text{CE}(f_{\text{add}(r, l^*, \alpha)}(p_{\text{false}}), t_{\text{true}})$ 
13:    $\mathcal{L}_{\text{ret}} \leftarrow \text{KL}(f_{\text{ablate}(r)}(p_{\text{retain}}) \parallel f(p_{\text{retain}}))$ 
14:   return  $\lambda_{\text{abl}} \mathcal{L}_{\text{abl}} + \lambda_{\text{add}} \mathcal{L}_{\text{add}} + \lambda_{\text{ret}} \mathcal{L}_{\text{ret}}$ 
15: end function
```

collapsing onto co-linear directions. In practice, activation-space directions cannot be scaled arbitrarily without model degeneration, which effectively bounds λ_i .

A naive extension of TDO would optimize only the basis vectors, but this causes *basis collapse*: the loss is satisfied at each v_i while interior directions of the cone remain unconstrained. I avoid this by evaluating the composite loss both on the basis vectors and on Monte-Carlo samples drawn from the cone interior at every step. Computing the loss on the basis vectors improves stability and tightens the lower bound of the ASR, since the basis vectors form the cone boundary and tend to degrade first. After each gradient step, I re-project the basis onto the cone via modified Gram–Schmidt orthonormalization, which restores orthonormality of V .

Sampling the interior requires care. Directional ablation depends only on the normalized $\hat{r}$, so what matters is the *direction* of the sample, not its magnitude. A natural choice would be to draw non-negative coefficients λ_i , form a convex combination $r = \sum_i \lambda_i v_i$, and normalize; but the map from coefficients to normalized directions is uneven, and most such samples cluster near the basis vectors after normalization, biasing the estimate toward the cone boundary. I instead sample directions uniformly from the non-negative orthant of the unit sphere in the cone’s coordinate space, exploiting the rotational symmetry of the isotropic Gaussian: at each step, 16 directions $\{u_j\}$ are drawn by sampling $c_j \sim \mathcal{N}(0, I_k)$, taking component-wise absolute values to fold the mass into the positive orthant, and setting $u_j = |c_j| / \| |c_j| \|_2$. Each $u_j \in \mathbb{R}^k$ is a unit vector of non-negative coefficients expressed in the basis of the cone; the map $r = V u_j$ realizes it as a direction in the model’s residual stream $\mathbb{R}^d$, where the intervention is applied. This Gaussian-based scheme yields uniform coverage of the cone and avoids the boundary bias of convex-combination sampling (Wollschläger et al. 2025).

Algorithm 2 Truth Cone Optimization (TCO)

Require: Frozen model f , scaling coefficient α , addition layer l^* , learning rate η , loss weights $\lambda_{\text{abl}}, \lambda_{\text{add}}, \lambda_{\text{ret}}$, cone dimension k , and data $\mathcal{D} = \{(p_{\text{true},i}, p_{\text{false},i}, p_{\text{retain},i})\}_{i=1}^N$.

Ensure: Orthonormal cone basis $V = [v_1, \dots, v_k]$

```
1: if  $k = 1$  then
2:    $V \leftarrow \theta_{\text{DIM}} / \|\theta_{\text{DIM}}\|_2$ 
3: else
4:    $V \leftarrow [V^{(k-1)}, v_{\text{rand}}]$  ▷ augment previous basis
5:    $V \leftarrow \text{GRAMSCHMIDT}(V)$ 
6: end if
7: while not converged do
8:   Sample batch  $\mathbf{B} \sim \mathcal{D}$ 
9:    $\mathcal{L}_{\text{sample}} \leftarrow \mathbb{E}_{u \sim \text{SAMPLE}(V)} [\text{COMPUTELOSS}(u, f, \mathbf{B})]$ 
10:   $\mathcal{L}_{\text{basis}} \leftarrow \frac{1}{k} \sum_{i=1}^k \text{COMPUTELOSS}(v_i, f, \mathbf{B})$ 
11:   $\mathcal{L} \leftarrow \mathcal{L}_{\text{sample}} + \mathcal{L}_{\text{basis}}$ 
12:   $V \leftarrow V - \eta \nabla_V \mathcal{L}$ 
13:   $V \leftarrow \text{GRAMSCHMIDT}(V)$  ▷ re-orthonormalize
14: end while
15:
16: function  $\text{SAMPLE}(V)$ 
17:    $s \sim \text{Unif}(x \in \mathbb{R}_+^k : \|x\|_2 = 1)$ 
18:    $r = Vs$ 
19:   return  $r$ 
20: end function
```

Initialization is incremental: for $k = 1$, I set V from θ_{DIM} as in TDO; for $k > 1$, I augment the previous $(k-1)$ -cone basis with one random unit vector and re-orthonormalize, adding a single degree of freedom per increment. All other hyperparameters are inherited from TDO (Table 6). Algorithm 2 gives the full procedure.

2.6 Evaluation protocol

Two evaluations check whether a learned cone captures *novel* structure and whether it does so without damaging general capabilities.

Geometric alignment with DIM.

A high-ASR cone that merely replicates the DIM direction k times would be trivial. For each cone of dimension $k \in \{1, 2, 3, 4\}$ I compute the cosine alignment $|\cos(\theta_{\text{DIM}}, v_i)|$ for every basis vector and a cone adds novel structure if $|\cos(\theta_{\text{DIM}}, v_i)| \approx 0$ for multiple v_i while each direction stays individually causally effective (Table 4).

KL retention on filtered Alpaca.

High ASR result is useless if the intervention destroys general capabilities. For each (model, k) pair I sample 32 directions from the cone interior (Algorithm 2) and compute the mean KL divergence between the ablated and clean model over the last 30 token positions of 100 held-out Alpaca instructions filtered of factual-recall keywords. A cone is *surgical* if the mean MC KL falls below 0.1 (Arditi et al. 2024), about 96,000 token-level comparisons per cone.

3 Experiments

Table 1 Cross-model summary. l^* : target hidden-state layer from Exp 1. $\alpha = \|\theta_{\text{DIM}}\|_2$: intervention magnitude. ASR_{abl} : in-distribution Answer Switching Rate under ablation. “Surgical” columns mark whether mean KL < 0.1 on the retain set. Higher ASR ($\uparrow$) and lower KL ($\downarrow$) are better. The three models that fail *both* axes at $k = 1$ (red row tint) all recover *both* at $k = 2$.

Model	l^*	α	DIM $\text{ASR}_{\text{abl}} \uparrow$	TDO $\text{ASR}_{\text{abl}} \uparrow$	$k=2$ $\text{ASR} \uparrow$	$k=2$ KL $\downarrow$	Surg. $k=1$	Surg. $k=2$
Qwen-2.5-1.5B-Instruct	17	13.05	0.010	1.000	1.000	0.008	✓	✓
Gemma-2-2B-it	15	84.52	0.216	1.000	1.000	0.004	✓	✓
Qwen-2.5-7B-Instruct	18	26.61	0.000	0.020	1.000	0.037	×	✓
Gemma-2-9B-it	24	106.56	0.000	0.774	1.000	0.001	✓	✓
Llama-3.1-8B-Instruct	14	4.07	0.020	1.000	1.000	0.017	×	✓
Qwen-2.5-14B-Instruct	30	48.25	0.000	0.658	1.000	0.009	×	✓

3.1 Experimental setup

Models.

Six instruction-tuned models from three families: Qwen-2.5 (1.5B, 7B, 14B) Yang et al. (2024), Gemma-2 (2B, 9B) Gemma Team (2024), and Llama-3.1-8B Grattafiori et al. (2024). All are loaded via nnsight Fiotto-Kaufman et al. (2025) on a RunPod NVIDIA A40 48GB instance.

Datasets.

Three True/False datasets: **cities** (“The city of [City] is in [Country]”, 1 496 statements) Marks and Tegmark (2024), **animals** (“A [Animal] is a mammal”, 504), **elements** (“The symbol for [Element] is [Symbol]”, 930), plus a **retain** set of 100 filtered Alpaca instructions. Each factual dataset is split 80/20 for ID evaluation; OOD evaluation trains on cities and animals and evaluates on elements.

Metrics.

ASR_{abl} , ASR_{add} , and KL retention (Section 2.1); logit difference for localization (Section 2.2). Hyperparameters in Table 6.

3.2 TDO improves on the DIM baseline (RQ1)

Research question: Does gradient-based 1-D optimization recover causally more precise directions than the Difference-in-Means baseline at matched layer and intervention magnitude?

Table 2 reports ASR_{abl} and ASR_{add} for both methods, ID and OOD.

- **TDO closes the DIM ablation gap on five of six models.** For Qwen-2.5-1.5B, Gemma-2-2B, Gemma-2-9B, Llama-3.1-8B, and Qwen-2.5-14B, TDO ablation ASR moves from ≤ 0.216 (DIM) to between 0.658 and 1.000. The composite loss optimizes $\mathcal{L}_{\text{abl}}$ directly, while DIM optimizes only mean separation. The OOD numbers suggest TDO is not memorizing entity-specific noise: TDO ablation OOD stays ≥ 0.125 on every model except Qwen-2.5-7B.
- **TDO at $k = 1$ fails for Qwen-2.5-7B.** Here both DIM and TDO give near-zero ablation ASR (0.000 and 0.020). This is a clear failure of the 1-D method on one model and motivates Section 3.3, where $k = 2$ reaches $\text{ASR} = 1.000$ ID for it.

Table 2 Exp 1 — Truth Direction Optimization (TDO) versus Difference-in-Means (DIM) at $k = 1$. ID: in-distribution (train/val split per source). OOD: train on two of {cities, animals, elements}, evaluate on the held-out third. **Best** per (model, intervention, split). The single TDO failure case is Qwen-2.5-7B (italicized): at $k = 1$, neither method achieves meaningful ablation ASR; this motivates Section 2.5.

Model	In-distribution (ID)				Out-of-distribution (OOD)			
	DIM abl	TDO abl	DIM add	TDO add	DIM abl	TDO abl	DIM add	TDO add
Qwen-2.5-1.5B	0.010	1.000	1.000	1.000	0.133	0.998	1.000	1.000
Gemma-2-2B	0.216	1.000	1.000	1.000	0.071	1.000	1.000	1.000
<i>Qwen-2.5-7B</i>	0.000	0.020	0.995	0.940	0.015	0.080	0.974	0.813
Gemma-2-9B	0.000	0.774	0.714	0.960	0.045	0.327	0.837	0.963
Llama-3.1-8B	0.020	1.000	1.000	1.000	0.185	1.000	0.985	1.000
Qwen-2.5-14B	0.000	0.658	1.000	1.000	0.011	0.125	1.000	1.000

3.3 The 2-D Truth Cone closes the in-distribution gap (RQ2)

Research question: Does extending the optimization to a k -dimensional cone (with Monte-Carlo interior sampling) close the residual ASR gap where 1-D fails, and at what k ?

Figure 1(a) and Table 3 report mean Monte-Carlo Answer Switching Rate over 32 directions sampled uniformly from each learned cone, ID and OOD, for $k \in \{1, 2, 3, 4\}$.

- **TCO at $k=2$ saturates ID ASR for every model.** All six reach mean MC ASR = 1.000 in-distribution at $k = 2$, including the Qwen-2.5-7B failure case. One extra dimension is enough to bridge the 1-D gap in-distribution across the three families.
- **MC interior sampling gives uniform mediation.** The MC standard deviation at $k \leq 2$ is zero or near-zero for all models (Table 3). Because the basis is kept orthonormal and the loss is applied to 16 interior samples per step, the interior has no “dead zones” an ablation could find by chance. The MC std grows only at $k = 4$ for Qwen-2.5-1.5B and Llama-3.1-8B, where over-parameterization starts to leave interior pockets unconstrained.

- **OOD generalization is weakest for Qwen-2.5-7B.** Its OOD ASR at $k = 2$ is 0.682, below the ID 1.000 but rising to ≥ 0.987 at $k = 3$. This model needs more degrees of freedom to generalize across domains even though $k = 2$ suffices within-domain; I return to this in the limitations.

Table 3 Exp 2 — Mean MC ASR (ablation) over 32 samples drawn from the learned cone. For all six models, $k = 2$ suffices to reach $\text{ASR} = 1.0$ in-distribution. The standard deviation column (ID only, $k = 2$) is 0 for all models, indicating uniform mediation across the cone interior. k^* is the smallest k at which the model is simultaneously surgical (Tab. 5) and saturated on ASR.

Model	In-distribution (ID)				Out-of-distribution (OOD)				k^*
	$k=1$	$k=2$	$k=3$	$k=4$	$k=1$	$k=2$	$k=3$	$k=4$	
Qwen-2.5-1.5B	1.000	1.000	1.000	1.000	0.998	0.998	0.999	0.987	2
Gemma-2-2B	1.000	1.000	1.000	1.000	1.000	1.000	0.997	0.939	2
Qwen-2.5-7B	0.020	1.000	1.000	1.000	0.080	0.682	0.987	0.993	2
Gemma-2-9B	0.774	1.000	1.000	1.000	0.327	1.000	0.980	0.946	2
Llama-3.1-8B	1.000	1.000	0.807	0.983	1.000	1.000	0.916	1.000	2
Qwen-2.5-14B	0.658	1.000	1.000	0.994	0.125	0.998	0.938	0.894	2

3.4 The cone basis is near-orthogonal to the linear probe (RQ3)

Research question: Are the cone’s basis vectors aligned with the linear DIM probe, or do they capture structure the linear probe does not represent?

Table 4 reports $|\cos(\theta_{\text{DIM}}, v_i)|$ at the reference cone dimension $k = 4$ for every model.

- **At $k = 4$, the entire basis is near-orthogonal to DIM.** For all six models and all four basis vectors, $|\cos(\theta_{\text{DIM}}, v_i)| < 0.1$, most below 0.05. Since these same cones reach mean MC ASR = 1.000 (Table 3), they are causally effective directions the DIM probe does not recover.
- **DIM-alignment is specific to $k = 1$.** At $k = 1$, $|\cos(\theta_{\text{DIM}}, v_1)|$ sits between 0.816 and 0.932 (right column): the warm-started direction stays close to DIM. This does not persist. As k grows the optimizer re-fits the whole basis, and by $k = 4$ even v_1 has rotated to near-orthogonal. The cone does not keep a DIM-aligned axis; it uses DIM only as a starting point.

Together, the high MC ASR and the near-zero cosine similarity across the full basis make the “ k noisy copies of DIM” explanation unlikely: the cone mediates truth while its basis is geometrically distinct from the linear probe.

3.5 Surgicality: $k=2$ resolves the 1-D KL failure (RQ4)

Research question: Do cone interventions preserve unrelated capabilities, and how does surgicality behave as a function of k ?

Figure 1(b) and Table 5 report mean Monte-Carlo KL divergence on 100 filtered Alpaca prompts, 32 MC directions per cone, 30 token positions per prompt.

Table 4 Exp 3 — $|\cos(\theta_{\text{DIM}}, v_i)|$ at reference $k = 4$. At $k=1$ the single basis vector stays near-aligned with DIM (right column, warm-start sanity check), but in the full 4-dimensional cone *every* basis vector — v_1 included — has $|\cos| < 0.1$ for all six models. DIM-alignment is therefore specific to $k=1$: as the cone grows the whole basis rotates into structure the linear probe does not capture.

Model	$ \cos(\theta_{\text{DIM}}, v_i) $ at $k = 4$				$ \cos $ at
	v_1	v_2	v_3	v_4	$k = 1$
Qwen-2.5-1.5B	0.006	0.030	0.019	0.050	0.932
Gemma-2-2B	0.031	0.096	0.012	0.070	0.905
Qwen-2.5-7B	0.012	0.063	0.061	0.011	0.868
Gemma-2-9B	0.084	0.017	0.017	0.009	0.860
Llama-3.1-8B	0.088	0.037	0.010	0.041	0.842
Qwen-2.5-14B	0.060	0.005	0.044	0.021	0.816

- **Three of six models are not surgical at $k = 1$.** Qwen-2.5-7B, Qwen-2.5-14B, and Llama-3.1-8B exceed the threshold $\text{KL} = 0.1$ under 1-D ablation, at 0.292, 0.194, 0.282. The other three pass at $k = 1$.
- **All six are surgical at $k = 2$.** Every model’s mean MC KL drops below 0.05; the three non-surgical models recover to 0.037 (Qwen-2.5-7B), 0.009 (Qwen-2.5-14B), and 0.017 (Llama-3.1-8B).
- **Co-failure and co-recovery.** The three models that fail surgicity at $k = 1$ are exactly the three that fail ASR_{abl} at $k = 1$ (Sections 3.2 and 3.3), and both axes recover at $k = 2$. A 1-D projection of a multi-dimensional structure would both under-specify the concept (low ASR) and disturb correlated orthogonal components (high KL), which matches the joint failure seen here: for the larger models a single direction appears mismatched to truth on both causal axes at once, not under-expressive in just one.
- **Higher k can re-introduce non-surgicity.** At $k = 3$, Llama-3.1-8B and Qwen-2.5-7B rise back above the threshold (0.811 and 0.202). This points to $k = 2$ as the safe operating point rather than “higher is always better”.

The KL term $\mathcal{L}_{\text{ret}}$ is applied to every Monte-Carlo-sampled direction, so the optimizer cannot satisfy the ablation loss along directions that destroy retain behaviour. The cone is therefore pushed toward a region where ablation is both effective and surgical, which a 1-D direction cannot guarantee for the larger models.

3.6 Cross-experiment summary

Table 1 consolidates the per-model results. Three observations.

- **$k = 2$ is the effective dimensionality for most models.** For five of six the best k^* is 2; the smallest model (Qwen-2.5-1.5B) is essentially flat across k and could equally be called $k^* = 1$. I do not claim d^* scales continuously with model size; the data fit a threshold reading, below about 7B parameters 1-D suffices, at or above 7B a 2-D cone is needed.

Table 5 Exp 4 — Mean Monte-Carlo KL divergence on the retain set. 32 directions per cone $\times$ 100 Alpaca prompts $\times$ last 30 token positions. Bold + light-red highlighting marks non-surgical entries ($\text{KL} \geq 0.1$, Arditì et al. 2024). The same three models that fail ASR at $k=1$ (Tab. 2) also fail surgicity there, and all six pass at $k=2$.

Model	$k=1$	$k=2$	$k=3$	$k=4$	All surg.?
Qwen-2.5-1.5B	0.042	0.008	0.009	0.008	✓
Gemma-2-2B	0.028	0.004	0.007	0.012	✓
Gemma-2-9B	0.048	0.001	0.001	0.002	✓
Llama-3.1-8B	0.282	0.017	0.811	0.017	×
Qwen-2.5-14B	0.194	0.009	0.027	0.010	×
Qwen-2.5-7B	0.292	0.037	0.202	0.044	×

- **Consistent across the three families tested.** The same $k^* = 2$ holds for Qwen-2.5, Gemma-2, and Llama-3.1. The absolute layer l^* varies (Table 1), and Llama-3.1-8B is an outlier at $l^* = 14/32$ (depth 0.44), but the cone-dimension finding is consistent across families.
- **Llama-3.1-8B’s outlier l^* lines up with its $k = 1$ non-surgicity.** Its earlier l^* may explain the 1-D non-surgicity: ablating an earlier hidden state cascades through more downstream computation and may touch unrelated capabilities. A mechanistic account is beyond this paper’s scope and is a natural follow-up.

4 Conclusion

I asked whether the concept-cone framework that explains refusal transfers to truth, and at what effective dimensionality. For the models I tested, the evidence is consistent with a 2-D cone. Gradient-based 1-D optimization (TDO) recovers truth directions that are more causally precise than the DIM baseline (Table 2); extending it to a k -dimensional cone with Monte-Carlo interior sampling (TCO) reaches mean MC Answer Switching Rate of 1.0 in-distribution for every model at $k = 2$ (Table 3), with the cone interior, not only its basis, appearing to mediate truth. The added basis vectors v_2, v_3, v_4 are near-orthogonal to the DIM probe yet each is individually causally effective (Table 4), so the cone seems to capture components of factual representation the linear probe misses. The clearest single result is the joint pattern in Table 5: the three larger models that fail the ablation axis at $k = 1$ are exactly the three that fail surgicity at $k = 1$, and all three recover both axes at $k = 2$. This cone is a higher-dimensional *linear* object: every direction I recover is a linear direction in the residual stream, and the single-direction picture is its $k = 1$ case. The goal was to extend the linear representation hypothesis to a multi-dimensional regime for truth.

Limitations.

The empirical claims are scoped to instruction-tuned models on forced-choice True/False prompts over three factual domains (geography, biology, chemistry). Truth in open-ended generation, multi-hop reasoning, base (non-instruction-tuned) models, and

adversarial robustness of the cone under steering attacks remain untested. While six models across three families provide reasonable breadth, they are all relatively small ($\leq 14\text{B}$ parameters); whether the $k^* = 2$ threshold persists, shifts, or grows for frontier-scale models ($\geq 70\text{B}$) is an open question.

A likely contributor to the surgicity regressions observed at $k \geq 3$ is the fixed Monte-Carlo sampling budget: only 16 interior directions are drawn per training step (Table 6). As the cone dimension grows, the volume of the positive orthant of S^{k-1} increases and 16 samples may no longer cover it densely enough to prevent unconstrained interior pockets, precisely the “dead zones” the sampling is designed to eliminate. Running TCO with a larger sampling budget (e.g. 64 or 128 directions per step) could improve higher- k surgicity, but was not feasible within the compute budget available for this work (single-GPU training on a rented instance). Each (model, k) configuration is also a single training run; variance across random initializations of the augmentation vector is not characterized.

OOD generalization at $k = 2$ is weakest for Qwen-2.5-7B (0.682, Table 3), so $k = 2$ as an effective dimensionality should be read as “in-distribution sufficient and OOD-acceptable on most models” rather than a universal mechanistic claim.

Future work.

Three directions follow. First, the basis vectors $v_1, \dots, v_k$ are causally effective and orthogonal to the linear probe, but their semantic content is unknown; logit-lens decoding and sparse-autoencoder analysis could test whether they match recognizable sub-concepts (e.g. entity type, factual domain). Second, an input-space accessibility analysis, in the spirit of Wollschläger et al. (2025)’s GCG suffix study, would show whether these directions are actually used by the model on specific prompts or are latent. Third, extending the framework to open-ended factual generation and base models would test whether the cone structure observed here is a property of instruction tuning or of the underlying representations.

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A Hyperparameters

Table 6 Hyperparameters for TDO and TCO. Inherited from Wollschläger et al. (2025) Table 3 and held fixed across all models.

Component	Value
Effective batch size	16 (gradient accumulation, physical batch 1)
Base learning rate	10^{-2}
LR schedule	plateau-reduce $\times 0.1$, patience 5, max 2 reductions
Optimizer	AdamW, weight decay 0
Gradient clipping	max-norm 1.0; spherical projection (TDO only)
λ_{abl}	1.0
λ_{add}	0.2
λ_{ret}	1.0
α (intervention magnitude)	$\ \theta_{DIM}\ _2$ per model
MC samples / accumulation step	16
Effective MC samples / batch	32
Cone re-orthogonalization	modified Gram-Schmidt every step
KL surgicality threshold	0.1
Retain set size	100 Alpaca prompts (factual-keyword filtered)
KL evaluation positions	last 30 tokens per prompt

AI Usage Disclaimer.

Parts of this project have been developed with the assistance of Google's Gemini Pro. The AI was used to support code refactoring (restructuring code for clarity and modularity), library understanding (learning how the **nnsight** tracing API works and how to correctly intervene on model forward passes), experiment design (brainstorming the pipeline structure and evaluation protocol), debugging (diagnosing runtime errors and understanding unexpected model behaviour), and report writing (drafting and refining sections of the written report). All AI-generated suggestions have been reviewed, understood, and adapted by me. The research question, methodology, implementation, experimental execution, result collection, and their interpretation are entirely my own work. I take full responsibility for the final content, its accuracy, and its academic integrity.